



FOOD AND NUTRITION

6905/01

Specimen -October/November 2027-2029

2 hours

Total: 100 Marks

Confidential

MARK SCHEME

SECTION A

1 (a) (i) Name monosaccharides that form Maltose.

Glucose + glucose $\longrightarrow$ maltose [2]

(ii) Name other monosaccharide.

Galactose

Fructose (Any one) [1]

(b) State one food that provides protein:

(i) Zein - Maize [1]

(ii) Gliadin - Wheat [1]

(c) Explain the term essential Amino Acids.

- Amino Acids that the body cannot make - must be obtained from foods containing proteins in the diet. [2]

(d) State one reason why the body needs the following:

(i) Iron

- to give blood its red colour
- It is required to transport oxygen around the body to every cell
- formation/manufacture of haemoglobin in the blood;
- formation/manufacture of myoglobin in the blood;
- efficient functioning of enzymes in the body cells;
- works with enzymes to release energy from food;
- prevents anaemia;

(any one point) [1]

(ii) Chloride

- help in the formation of gastric juices/ production of hydrochloric acid.
- regulates pH balance for body fluids/maintains correct concentration of the body fluids.

(Any one point) [1]

2. (a) State three ways in which Vitamin B2 can be destroyed during cooking:

- Riboflavin is soluble in water/ it can be dissolved in water.
- Destroyed if heated in the presence of an alkali.
- Exposure to light also destroys the vitamin.
- Destroyed if overcooked.

(Any three points) [3]

(b) (i) **State the reason why extra water is essential in lactation.**

- Facilitates milk production.
- Additional fluid is needed to make up for the loss in the body.
- Quenches the thirst.

(Any one)

[1]

(ii) **State signs of dehydration.**

- pale
- very dry skin
- feeling dizzy
- sunken eyes
- rapid breathing
- headache
- not peeing or very dark peeing
- lack of energy – sleepiness
- irritability
- dry mouth/increased thirst.

(Any three)

[3]

(c) **Describe the differences between Anorexia Nervosa and Marasmus.**

Anorexia	Marasmus
Diet-related disease, which mainly affects teenage girls and young women	Diet-related disease, which mainly affects children under one year of age
Caused by a lack of appetite or sometimes self-consciousness about losing weight or self-starvation	Caused by a lack of food or starvation
Difficulty in eating normally, and the body rejects food	The body adapts to the shortage of food by depletion of fat stores

(Any two points for each condition)

[4]

(d) **Explain the effects of taking a highly refined diet.**

- **NSP content is removed**, which is responsible for absorbing water to make faeces/ stools soft or bulky, thus making the stools hard.
- **Hard stools lead to bowel disorders** - constipation/ diverticular disease/ piles/ colon cancer.
- **Leads to obesity** - most of them contain high amounts of sugar, which, when stored in the body, is converted to fat/highly refined foods lack NSP, which regulates cholesterol levels.

(Any two well-explained points)

[4]

3. (a) (i) **Calculate the Body Mass Index (BMI) of a person who weighs 60 kg and has a height of 1.5 metres. Include the units.**

M1 Correct equation for calculation of BMI = Weight /height ² OR 60kg/1.5m²

M2 Evaluation of BMI= 26.7kg/m²

M3 Correct Units kg/m²

[3]

- (ii) **State the weight category this person falls into based on the calculated BMI value in (a)(i).**

Overweight

(Error carried forward (ECF) from 3(a)(i). Overweight should only be credited if it matches the BMI calculation in (i). **[1]**

- (b) State the energy value of 1g protein, 1g carbohydrate, and 1g fat.**

1g Protein = 4kcal/18.72kJ

1g Carbohydrate = 3.75kcal/4 kcal/18.72kJ

1g Fat = 9 kcal/37.62 kJ

[3]

- (c) State three functions of potassium in the body.**

- Potassium maintains the correct concentration of body fluids
- Facilitates the removal of excess sodium in the body
- Helps to prevent High BP
- necessary for muscle contractions and nerve signals

(Any three points)

[3]

- (d) Describe how mucus and muscular walls assist in the digestive system.**

- (i) **Mucus** lubricates the food and the membranes/lining of the digestive system
- (ii) **Muscular walls** of various parts of the digestive system push food along by regular contractions/ peristalsis

[1]

[1]

- (e) Explain the importance of the following in the digestion of food:**

- (i) Gall bladder**

Stores bile produced by the liver - which is needed for the emulsification in the duodenum.

[2]

- (ii) Gastric juice**

- A digestive juice that provides an acidic pH, which activates enzymes in the stomach
- The acidic environment kills most bacteria and pathogens in food

(Any one well-explained point)

[2]

[Total: 40 Marks]

SECTION B

4 (a) Name gases that raise flour mixtures.

- Oxygen
- Carbon dioxide
- Steam/Water vapour

(Any two- one mark each)

[2]

(b) (i) State the reason for the pastry to be baked at a high temperature until set.

- Prevents fat from running out of the pastry.

[1]

(ii) Describe changes during the cooking of pastry.

- Fat melts and some is absorbed into the flour, and some drains out of the pastry.
- Liquid turns into steam, which expands.
- Starch in the flour takes up water and gelatinise.
- Gluten coagulates to give a firm structure to the product.
- Surface starch forms dextrin.

(Any three points)

[3]

(c) State the effect of using the following ingredients when making bread rolls.

(i) Liquid with a temperature above 40°C

- temperature will destroy /kill the yeast plant and fermentation will not take place

[1]

(ii) Too little yeast

- the yeast mixture will not rise/ fermentation will be delayed

[1]

(iii) Too much fat

- Bread will not rise well/ yeast will be dormant
- Too much fat inhibits the gluten development

(Any one)

[1]

(d) Name three kitchen areas that make up the work triangle in the kitchen.

- Stove- cooking area
- Fridge-storage area
- Sink-washing area

[3]

(e) (i) State the factor that determines the difference in consistency of batters.

The amount of liquid used

[1]

(ii) Name the dishes prepared using the following batters:

Pouring batter: Toad- in- the - hole/ Yorkshire pudding/ Pancakes**Coating batter:** deep-fried fish/deep-fried chicken, fruit fritters

(Any one for each)

[2]

(f) (i) **State which group produced the cake with the best quality.**

Group A's cake

(ii) **Identify two different mistakes the learners from each of the other two groups might have made.**

Group B: oven temperature too high

Overbaking

Too much baking powder/ not mixing properly

Group C: Undercooked or low oven temperature

Too much liquid or fat

Insufficient raising agent

Overmixing

(Any two for each group)

[4]

[Total: 20 Marks]

5. (a) State rules to follow when preparing tough cuts of meat.

- Use slow moist heat methods/braise/stew
- Use the correct amount of liquid for the meat to be stewed
- Use a saucepan with a tight-fitting lid
- Cut meat across the grain into neat, moderate-sized chunks
- Tenderise before cooking/marinate with acidic ingredients
- Stew must not boil but simmer gently
- Use a heavy-based saucepan

(Any three points)

[3]

(b) (i) Give examples of TVP.

Soya mince, soya chunks, soya granules

(Any two)

[2]

(ii) Mention examples of dishes where TVP can be used instead of meat.

- Spaghetti bolognese, stews, soups, vegetable rolls, patties, curries

(Any two)

[2]

(c) (i) State one example of a traditional dish prepared using pulses.

Cow peas stew/ *sishwala/ siphuphe, umngcushu*

(Any one)

[1]

(ii) State the advantages of using a pressure cooker when cooking pulses.

- Less attention required for cooking.
- Saves time.
- Uses less fuel - it is therefore economical and saves money spent on fuel.
- Softens tough pulses easily.
- Little loss of nutrients- therefore nutritive value of food is maintained.

(Any three points)

[3]

(d) Discuss the responsibilities of the consumer when buying perishable food.

- Checking expiry/best before date - expired food may pose health risks.
- Inspect the appearance and smell/ look for signs of spoilage-they indicate poor quality.
- Avoid damaged packaging, which can lead to contamination.
- Buy suitable quantities, reduce waste, and ensure freshness.
- Read and follow food labels—preserve quality. **(Any two well-explained points) [4]**

(e) State the reason why wooden equipment should never be left to soak in water.

- Absorbs water and swells/cracks
- A damp environment encourages the growth of bacteria and moulds **(Any one) [1]**

(f) Explain two safe steps you should take to put out a small fire caused by burning cooking oil.

- Turn off the heat source-stop feeding the fire with more heat.
- cover the fire with a metal lid/damp cloth- to cut off oxygen.
- Use baking soda- smother the flame.
- Use a fire blanket- cuts off oxygen.

(Any two well-explained points) [4]

[Total: 20 Marks]

SECTION C

6. Discuss deep fat frying as a method of cooking food. Evaluate the health risks linked to this method of cooking [20]

Deep fat frying is done by completely immersing food in hot fat or oil in a deep fat fryer or saucepan, for example, when cooking chips.

Choice of fat or oil for deep-fat frying:

The fat must be suitable for heating up to 200 °C without burning, e.g., vegetable oils and lard. However, reduced-fat spreads have a high-water content and are not suitable for frying. Water causes fat to splatter when heated, and so it is dangerous.

Suitable Equipment:

Use a strong, deep pan, with a frying basket; electric fryers, with built-in thermostats, can be purchased for domestic use; they are safer as there is less risk of causing a fire.

Preparing food for deep-fat frying:

Food suitable for deep fried: e.g., fish, meat, fruit, should first be coated to prevent overcooking and the loss of juices from the food, and to prevent the food breaking up and absorbing too much fat. **Suitable coatings include:**

Beaten egg
 Beaten egg and breadcrumbs
 Beaten egg and seasoned flour
 Beaten egg and oatmeal
 Egg, flour, and oatmeal

Reasons for coating food: with batter, egg, and breadcrumbs

When the food is placed in the hot fat, the egg in the coating coagulates rapidly and thus forms a protective layer around the food, which becomes crisp and golden brown. The food inside continues to cook by conduction and retains its flavour and texture.

- Hold the shape of food/prevent breaking
- Prevent absorption of fat
- Protect food from heat/prevent burning
- Prevent loss of juices from food/ food cooks evenly

Advantages of frying: quick, adds caloric value without bulk, adds flavour, browns, and achieves a crisp texture.

Disadvantages of frying: More difficult to digest - Needs constant attention- more dangerous

Important rules for deep frying:

- Do not fill the pan more than halfway with oil or fat, as the oil or fat will rise rapidly when food is placed in it and could boil over.
- Lower the food gently into the hot fat. Do not drop it in, as it will splash and cause burns
- - Do not overfill the pan with food, may overflow and be difficult to turn, or considerably lower the temperatures of the fat and affect the finished result.
- Do not overheat, fat may ignite
- Use a dry pan- to prevent 'spitting' – as splashing oil causes burns.

- Turn the handles of the cooking pan in – so will not be knocked when passing.
- Do not overheat the fat- the outside of the food will cook quickly, and the inside will not cook sufficiently.
- Turn the food carefully to ensure even cooking
- Use absorbent kitchen paper to drain the food when cooked and a perforated spoon to lift the food out, or cooking tongs. A frying basket can also be used.
- When food is cooked, turn off the heat and allow fat to cool before straining it through a piece of muslin to remove impurities.
- Store fat in a cool, dry, dark place to prevent rancidity.

Safety rules for deep frying

- Never leave a frying pan unattended
- Do not heat the fat beyond the required temperature
- Keep the pan handle turned towards the side of the cooker to prevent it from being knocked over
- If the fat starts to smoke, turn the heat off immediately, as this is near its flash point

Fat temperature is too hot:

- Outside cooks quickly- inside raw- danger of food poisoning if food not thoroughly cooked- must reach 70°C - bitter flavour when overcooked fat temperature too low - Outside surface not sealed - protein coagulates - in egg - starch in flour - not gelatinised - as soon as food enters oil - oil absorbed by food - unappetising - difficult to digest.

Effect of heat/ when overheated:

Thin, bluish haze of smoke is given off- gives food an unpleasant flavour- reaches 'smoke point'. Wherein fat molecules split up, reducing the keeping quality of fat — ignites and burns fiercely, i.e., flash point.

Health problems that may be associated with frying:

- High fat and calorie content- fried foods absorb a significant amount of fat which can lead to weight gain and obesity
- risk of heart diseases-frequent consumption of fried foods is linked to increased levels of bad cholesterol, which can increase the risk of high blood pressure and heart attacks
- Digestive discomfort: Oily foods are difficult to digest for some people, may lead to bloating, gas, or indigestion
- Nutrient loss- high temperatures can destroy certain heat-sensitive vitamins, e.g, vitamin C, reducing the nutritional value of food
- Formation of harmful compounds- when oil is reused or overheated, it can break down and form trans fats, which are linked to cancers and other chronic illnesses

Band	Descriptors	Past Mark	Total
High	Candidate is able to: - Give a detailed understanding of deep fat frying (definition) - Give more examples of deep-fried food - Explain the most reasons for coating deep-fried food - Explain most benefits/advantages of deep-fat frying food - Give more points to consider when deep fat frying -have a clear understanding of safety rules to follow when deep fat frying - Have a clear understanding of the points to consider when deep-frying - Have a clear understanding of the health problems associated with deep-fat frying	14 – 20	20
Middle	Candidates may: - Show some understanding of deep fat frying (definition) - Give some examples of deep-fried food - Explain some mechanical functioning of a microwave oven - Explain some reasons for coating deep-fried food - explain some benefits/advantages of deep-fat frying food - Give some points to consider when deep fat frying - have some understanding of safety rules to follow when deep fat frying - Have some understanding of the points to consider when deep-frying - Have some understanding of the health problems associated with deep-fat frying	9 - 13	
Low	Candidates may: - Shows little understanding of deep fat frying (definition) - Give a few examples of deep-fried food - Explain a few reasons for coating deep-fried food - Explain a few benefits/advantages of deep-fat frying food - Give a few points to consider when deep-frying - Have little understanding of the safety rules to follow when deep-frying - Have little understanding of the points to consider when deep-frying - Have little understanding of the health problems associated with deep-fat frying	0 – 8	

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